$$g_B = \nabla_\theta L_B, \quad g_U = \nabla_\theta L_U$$

$$L_B(\theta - \eta(g_B + \lambda_U g_U)) - L_B(\theta)$$

$$= -\eta(\|g_B\|^2 + \lambda_U g_B^\top g_U) + \mathcal{O}(\eta^2)$$